

# Aman Yadav

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## SUMMARY

Mathematics & Computing undergraduate at Central University of Karnataka, focused on applied machine learning, deep sequence modeling, and automotive sensor telemetry. Experienced deploying low-latency PyTorch and FastAPI microservices for vehicle diagnostics and battery State of Health (SOH). Former engineering intern at Maruti Suzuki.

## EDUCATION

### Central University of Karnataka

Kalaburagi, India

*B.Tech in Mathematics & Computing (Expected 2027)*

- **Relevant Coursework:** Machine Learning, Deep Learning, Probability & Statistics, Linear Algebra, Optimization, Time-Series Analysis, Data Structures & Algorithms.

## TECHNICAL EXPERIENCE

### Maruti Suzuki India Limited – Engineering Intern (EN-DAX Dept)

Gurugram, India

*Applied ML & Vehicle Telematics*

*20 May 2026 – 15 July 2026*

- **Telemetry Pipeline:** Architected an industrial **6-stage pipeline** normalizing raw CAN-bus sensor feeds into a **40-feature automotive taxonomy** with automated trip segmentation.
- **Kinematics & Behavior ML:** Derived continuous Haversine trajectories and speed differentials to detect **aggressive driving maneuvers** (harsh braking, rapid acceleration).
- **Cleansing & Analytics UI:** Implemented **IQR/Z-score outlier filtering** with JSON audit trails; deployed a real-time **Streamlit console** for vehicle dynamics engineers.

## FEATURED PROJECTS

### EV-Lifespan — EV Battery Degradation & RUL Platform



- **Deep Sequence Architecture:** Engineered a 2-tier spatial-temporal model combining a **1D-CNN** (voltage/current/temperature profiles) with **stacked LSTMs** to predict battery Remaining Useful Life (RUL).
- **Accuracy Benchmark:** Achieved an **RMSE of 19.40 cycles** and **15.67 MAE cycles** with **<15ms latency** served via a high-throughput **FastAPI** microservice.
- **Containerized Dashboard:** Deployed in a multi-stage **Docker** environment streaming degradation curves and battery State of Health (SOH) indicators to an interactive UI.

### Car Price Predictor — Automotive Valuation Engine



- **Feature Engineering:** Built automated tabular preprocessing pipelines handling mileage, manufacturing year, and engine specs with **target encoding** and **leak-free transforms**.
- **Model Benchmarking:** Evaluated **Gradient Boosting** and **Random Forest** with 5-fold cross-validation; optimized hyperparameters to maximize  $R^2$  and minimize valuation variance.
- **Production Serving:** Deployed containerized model inference on **Render** for instant, real-time automotive market valuations.

### EduPulse — Predictive Analytics Platform



- **Dual-Model Architecture:** Designed predictive ML pipelines coupling baseline priors with longitudinal models using **GroupedKFold cross-validation** to prevent data leakage.
- **Ethical AI & Governance:** Enforced demographic quarantine in feature selection to eliminate algorithmic bias and ensure zero PII leakage.
- **Asynchronous Inference:** Scaled batch predictions using an asynchronous **Celery worker queue** backed by **Redis**, maintaining **< 40ms API response latency**.

## SKILLS

**ML & Deep Learning:** PyTorch, Scikit-Learn, 1D-CNN, LSTM, Time-Series Forecasting, Feature Engineering, K-Fold CV  
**Data Science & Telemetry:** Pandas, NumPy, Automotive CAN-Bus Telemetry, Signal Cleansing, Matplotlib, Seaborn  
**Backend & Deployment:** FastAPI, Django, REST APIs, Celery, Redis, Docker, Streamlit, Microservices  
**Languages & Foundations:** Python, C++, SQL (PostgreSQL), TypeScript, Linear Algebra, Probability & Statistics